Ethnic Diversity and Cooperation After Disaster: Evidence from Hurricane Harvey

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Abstract

Under what conditions does disaster exposure shape cooperation? We study how ethnic diversity moderates the relationship between disaster damage and cooperation. We argue that diversity shapes different forms of cooperation depending on the feasibility of monitoring. When monitoring is feasible, as in face-to-face interactions, we do not expect the relationship between disaster damage and cooperation to vary systematically with diversity. By contrast, when monitoring is more difficult, as in the provision of public goods such as disaster recovery and adaptation policies, we expect a diversity penalty that increases with disaster damage. We examine these predictions using observational and experimental data from Hurricane Harvey, which generated substantial variation in local exposure to disaster damage. Consistent with prior work, diversity is associated with weaker pre-disaster cooperation with neighbors. Following the storm, we find no diversity penalty in interpersonal, face-to-face helping behavior, but document that the diversity penalty in support for disaster recovery and adaptation policies increases with damage. These findings help reconcile mixed evidence on disaster exposure and cooperation by showing that this relationship varies with both the form of cooperation and local ethnic diversity.

Keywords: natural disasters; ethnic diversity; cooperation; public goods

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Introduction

Under what conditions does disaster exposure shape cooperation? A growing literature documents that disasters tend to increase prosocial behavior and that social capital facilitates recovery. However, the evidence remains mixed: some studies document positive effects, where exposure to natural disasters is associated with higher cooperation (Calo-Blanco et al., 2017; Cassar et al., 2017; Dacy and Kunreuther, 1969; Douty, 1972; Dynes, 1994; Méon and Verwimp, 2022; Toya and Skidmore, 2014; Schilpzand, 2023; Rao et al., 2011). Others report mixed or negative effects (Albrecht, 2018; Berrebi et al., 2021; Fleming et al., 2014; Rayamajhee et al., 2024). Much of this research treats disasters as uniform shocks, but whether shocks foster cooperation may depend on the social environment in which they occur.

We study how ethnic diversity, an important feature of the social environment, moderates the relationship between disaster damage and cooperation. Prior research suggests that shocks such as natural disasters foster cooperation (Méon and Verwimp, 2022) and that societies more exposed to such shocks have stronger social norms (Gelfand et al., 2011). Diversity, by contrast, tends to undermine cooperation and the provision of public goods (Alesina et al., 1999; Alesina and La Ferrara, 2000; Algan et al., 2016; Costa and Kahn, 2003; Habyarimana et al., 2007; Luttmer, 2001; Miguel and Gugerty, 2005). Yet little attention has been given to how these two factors interact.

Hurricane Harvey, the second-costliest disaster in U.S. history, generated substantial tract-level variation in the stakes of cooperation. Harvey was highly destructive—causing $125 billion in damages, 68 direct deaths, and over 100 indirect deaths (Blake and Zelinsky, 2018)—and struck one of the largest and most ethnically diverse metropolitan areas in the United States. Because both

flood exposure and ethnic diversity vary across neighborhoods, the setting allows us to examine whether the diversity penalty increases with damage exposure.

We argue that diversity shapes different forms of cooperation in distinct ways, depending on the extent to which contributions can be monitored. Monitoring is a central mechanism for mitigating asymmetric information and detecting free-riding, enabling groups to sanction defectors and sustain cooperation (Greif, 1993; Habyarimana et al., 2007; Miguel and Gugerty, 2005; Ostrom, 1990; Williamson, 1985). In face-to-face interactions—such as helping a neighbor—where actions are directly observable and the act of helping serves as a costly signal, we do not expect diversity to impose a penalty on cooperation. By contrast, when monitoring is difficult and sanctioning is limited—as in contributions to tax-funded public goods—we expect diversity to be associated with lower levels of cooperation. Such contributions are not directly observable and their benefits are diffuse, making individual contributions difficult to monitor; in such settings, the diversity penalty is more likely to emerge. Disaster recovery and adaptation policies—such as flood-control infrastructure or costly recovery programs—fit this logic because they are tax-funded and non-excludable, making them particularly vulnerable to free-riding.

We study patterns of cooperation following Harvey using observational and experimental data. We draw on three original survey waves fielded by the Hobby School of Public Affairs at the University of Houston between 2017 and 2020, covering four counties in the Houston metropolitan area. We combine these survey data with detailed administrative records to construct measures of ethnic diversity and damage at the census-tract level.

Our main finding is that the relationship between diversity and cooperation depends on the form of cooperation. We find no diversity penalty in interpersonal, face-to-face helping behavior—giving or receiving assistance. By contrast, for impersonal cooperation in the provision of public goods, tract-level storm damage is associated with greater support for disaster recovery and adapta-

tion policies in more homogeneous areas; in affected areas, support declines as diversity increases. We find a similar pattern in a conjoint survey experiment. Support for costlier policies—those requiring higher tax increases—is modestly higher among respondents personally affected by the storm. However, among affected respondents, support for costly policies declines in more diverse areas.

We also assess two alternative explanations. First, the diversity penalty may reflect visible fiscal costs rather than monitoring constraints: policies that require taxes or public spending may elicit lower support in diverse areas. Second, the penalty may reflect the time horizon of the policy: forward-looking mitigation policies, whose benefits are uncertain and delayed, may elicit lower support than policies addressing damage already incurred. Neither alternative accounts for the diversity penalty we document. Overall, the patterns we find are more consistent with our argument that monitoring capacity structures how diversity shapes different forms of cooperation.

Our findings contribute to an expanding literature on how natural disasters affect cooperation and prosocial behavior in three ways. First, we move beyond proxies such as trust and cohesion (Cassar et al., 2017; Toya and Skidmore, 2014; Calo-Blanco et al., 2017; Schilpzand, 2023) and use more direct measures of cooperation, distinguishing between interpersonal and impersonal cooperation. This focus also sets our study apart from recent work on disasters and politics that examines political attitudes (Arin et al., 2025; Frost et al., 2025; Visconti, 2022a) and electoral outcomes (Heersink et al., 2022; Masiero and Santarossa, 2021). Second, we show that disaster exposure is associated with support for disaster recovery and adaptation policies, and that this relationship varies with local diversity. This adds to recent research linking disaster exposure to policy support (Arias and Blair, 2024; Bechtel and Mannino, 2023; Bergquist and Warshaw, 2019; Visconti, 2022b) and showing that local disaster risk and cross-cutting distributive interests can soften partisan divides over climate policy (Cayton and Crisher, 2026). Whereas this literature empha-

sizes disaster risk and distributive interests, we focus on a different moderator: ethnic diversity. Third, we connect disaster politics to research on ethnic diversity and public-goods provision. Our findings complement evidence that racial attitudes predict support for government disaster spending (Gilens et al., 2025) by shifting attention from individual attitudes to local ethnic diversity. They also extend work showing that diversity can undermine public-goods provision by weakening shared norms, monitoring, and sanctioning (Habyarimana et al., 2007; Miguel and Gugerty, 2005). We show that this logic applies to post-disaster settings, where the stakes of cooperation are high: the diversity penalty is concentrated in costly and impersonal forms of cooperation, where monitoring is more difficult.

Disasters, ethnic diversity, and cooperation

Cooperation occurs when an individual incurs a cost to provide a benefit to others (Henrich and Henrich, 2007). Cooperative interactions can occur interpersonally—helping a friend or a stranger—or at larger scales—such as voting, recycling, or paying taxes. While cooperation in small-scale societies can be explained by kinship (McNamara and Henrich, 2017), large-scale cooperation in diverse societies remains a puzzle that motivates substantial theoretical and empirical work (Bowles and Gintis, 2011; Henrich and Henrich, 2007; Seabright, 2010).

Convergent lines of research show that common shocks, such as natural disasters, are associated with increases in prosocial behavior, including trust, social cohesion, and helping behavior (Calo-Blanco et al., 2017; Cassar et al., 2017; Dacy and Kunreuther, 1969; Douty, 1972; Méon and Verwimp, 2022; Rao et al., 2011; Schilpzand, 2023). Ethnic diversity is also associated with individuals' propensity to cooperate. A large literature shows that diversity hinders cooperation and the provision of public goods (Algan et al., 2016; Alesina and La Ferrara, 2000; Costa and Kahn, 2003; Luttmer, 2001; Miguel and Gugerty, 2005). These two forces are typically studied

separately. Yet shocks may condition the relationship between diversity and cooperation. Natural disasters raise the stakes of cooperation by increasing demand for help, recovery, and public goods, making them an important context in which to study how diversity shapes cooperation. In particular, post-disaster recovery may be more difficult in heterogeneous, loosely connected communities, where it is harder to form expectations about the behavior of others (Storr and Haeffele-Balch, 2012).

We propose that diversity shapes different forms of cooperation, depending on the scope for free-riding and the feasibility of monitoring. Monitoring facilitates cooperation across domains: it makes behavior observable so that deviations can be detected and punished, whether in common-pool resources (Ostrom, 1990), public goods provision (Habyarimana et al., 2007; Miguel and Gugerty, 2005), contractual exchange (Williamson, 1985), or long-distance trade (Greif, 1993). We argue that diversity moderates the relationship between shocks and cooperation when the cost of monitoring contributions is high and sanctioning is limited. In such situations, diversity dampens impersonal cooperation. By contrast, in interpersonal, face-to-face cooperative interactions, such as helping a neighbor, each action is a costly signal that others can observe directly. Because monitoring occurs during the interaction itself, the scope for free-riding is limited. In such interactions, we do not expect a diversity penalty.

Cooperation aimed at providing public goods follows a different logic. These contributions are generally not directly observable, are tax-funded, and generate diffuse benefits. Consequently, they are susceptible to free-riding. Disaster recovery and adaptation policies, such as flood-control infrastructure or costly recovery programs, are examples of such public goods. We therefore predict that, for cooperation aimed at providing public goods, the diversity penalty increases with damage.

Our argument differs from accounts that attribute the negative correlation between diversity and public-goods provision to divergent preferences across ethnic groups (Alesina et al., 1999;

Alesina and La Ferrara, 2000), as well as from accounts that emphasize individual racial attitudes in disaster-spending preferences (Gilens et al., 2025). We instead emphasize the observability of contributions and the feasibility of sanctioning. This yields two predictions:

- When monitoring is feasible, as in face-to-face helping, we do not expect a diversity penalty following disaster exposure. In particular, the relationship between damage and cooperation should not systematically vary with diversity.
- When monitoring is difficult and sanctioning is limited, we expect a diversity penalty that increases with disaster damage. Support for costly recovery policies should decline with diversity more sharply in more heavily damaged areas.

We examine these predictions using survey data and a policy conjoint experiment. A partner-choice experiment, reported in the Online Appendix (OA), provides supplementary evidence related to the monitoring logic.

Hurricane Harvey: A costly disaster in a diverse metropolitan area

The Houston metropolitan area is one of the largest and most ethnically diverse metropolitan areas in the United States, with more than 7 million residents. It has grown by 25% since 2010, an increase of 1.5 million people. The city of Houston is 44.5% Hispanic or Latino, 24.1% White, 22.1% Black or African American, and 6.7% Asian (U.S. Census Bureau, 2022). More than 28% of the city's residents are foreign-born. Diversity at the metropolitan level coexists with segregation at the neighborhood level: pooling race counts across the 971 census tracts in the study area gives a study-area-wide fractionalization index of 0.69 (the probability that two randomly selected residents belong to different ethnic groups), whereas the tract in which the average resident lives has

an index of 0.53. Tract-level diversity ranges from 0 to 0.76, with an interquartile range of 0.41 to 0.63 (Figure 1). These features make the Houston metropolitan area an important setting for studying how local diversity shapes cooperation and support for disaster recovery and adaptation policies.

In the United States, disasters are increasing in frequency and cost. Between 1980 and 2024, the country experienced 403 weather and climate disasters that exceeded $1 billion each in damages, with total costs exceeding $2.9 trillion.[1] Hurricane Harvey, a Category 4 hurricane, was the second-costliest disaster in U.S. history, with damages of $125 billion, 68 direct deaths, and over 100 indirect deaths (Blake and Zelinsky, 2018). It caused widespread damage due to extreme rainfall—the heaviest in U.S. history—totaling 27 trillion gallons, with some areas receiving over 50 inches (Blake and Zelinsky, 2018). It displaced 30,000 people and destroyed more than 200,000 homes and businesses.[2]

The response to Harvey spanned federal, state, and local actors, alongside large civilian efforts. Federal and state responses included the U.S. Coast Guard, the Federal Emergency Management Agency (FEMA), and the Texas National Guard. Local responders included the Houston Fire Department, the Houston Police Department, and Harris County Sheriff's marine teams. Congress initially approved $15.25 billion in federal disaster relief, followed by additional appropriations. Over 890,000 households registered for FEMA aid. The National Flood Insurance Program (NFIP) paid out more than $8.92 billion in claims. Beyond the governmental response, Harvey triggered grassroots efforts and volunteer operations. Civil society organizations spearheaded emergent response efforts (Kendra and Wachtendorf, 2016; Wachtendorf et al., 2018). Churches and faith communities played a prominent role, providing services including financial support and housing

[1] U.S. Billion-Dollar Disasters: 1980–2024

[2] National Centers for Environmental Information (2025).

assistance (Texas Interfaith Center for Public Policy, 2019). Local volunteers and groups—such as the civilian "Cajun Navy"—conducted boat rescues and assisted with evacuations.[3]

Data

Survey data. The Hobby School of Public Affairs at the University of Houston conducted three survey waves in Harris, Fort Bend, Brazoria, and Montgomery counties following Hurricane Harvey. The first wave, fielded by telephone in November–December 2017, interviewed 2,002 adults and oversampled areas expected to have flooded. The second wave, conducted by telephone in June–July 2018, included 1,073 respondents, of whom 572 had participated in the 2017 survey and 501 were drawn from a fresh sample. The third wave, conducted online in May–June 2020, surveyed 1,065 respondents with demographic quotas for county, age, race, and gender.[4] Across the three surveys, the 2017 and 2018 waves shared a phone-based design and an emphasis on flood-exposed areas, while the 2020 wave shifted to an online mode with quota sampling. The surveys provide the outcome measures, individual controls, and a self-reported damage measure. OA Section A2 provides details on sampling and survey logistics, and OA Table A1 provides descriptive statistics for the survey variables used in the analysis.

Administrative data. To measure ethnic diversity and storm damage, we employ two administrative datasets: the 2015 American Community Survey (ACS) five-year estimates and the City of Houston Open Data website.

1. ***Ethnic diversity***. We calculate measures of ethnic diversity using data from the 2015 ACS. The ACS provides social and demographic data at the census tract level. We use ACS vari-

[3] "In all-hands-on-deck response to Harvey, lessons learned from earlier storms," *The Christian Science Monitor*, August 28, 2017.

[4] Survey weights were constructed separately by wave. The 2017 weights corrected the owner–renter imbalance in Harris County; the 2018 weights adjusted for owner–renter balance and racial composition across the four counties; and the 2020 weights adjusted for race, sex, age, and party identification.

ables that capture the number of individuals in each ethnic group in each tract: White or Anglo, Black or African American, Hispanic or Latino, Asian, and Other. We construct a standard measure of ethnic fractionalization, defined as the probability that two randomly selected individuals in a given tract belong to different ethnic groups.[5]

2. ***Storm damage***. We obtained tract-level measures of damage from the City of Houston Open Data website. The data contain count variables for housing units affected by Harvey, based on a range of assessment sources, including 311 service request calls, 911 incident calls, Solid Waste Management Department debris pickup locations, high water mark data obtained by the Public Works Department Flood Plain Management Office, FEMA individual assistance requests, and data from the National Flood Insurance Program. We construct a continuous tract-level measure of storm damage, defined as the share of housing units in each tract recorded as affected. The City of Houston compiled these records only within city limits.[6] The distribution is right-skewed: the median tract has 9 percent of its housing units affected, while a small number of flood-prone tracts reach much higher shares (OA Figure A1). Figure 1 maps tract-level ethnic diversity and self-reported Harvey damage by ZIP code across the study area.[7]

3. ***Tract-level controls.*** We use six tract-level controls drawn from the ACS 2015 five-year estimates: median household income, share of renter-occupied units, share of foreign-born residents, share of residents below the poverty line, share of residents who moved in the past year, and population density. These controls address the concern that diversity may proxy for

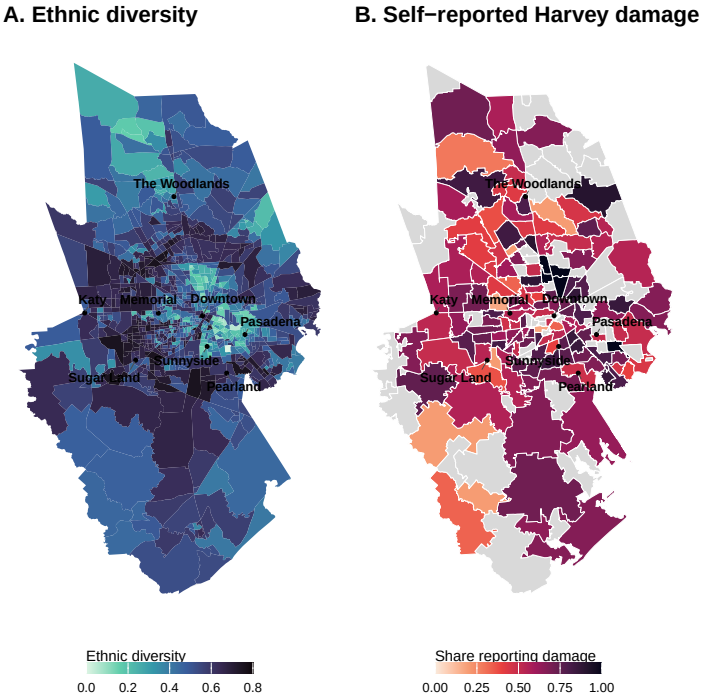
[5]Formally, ethnic fractionalization is defined as $1 - \sum_{g=1}^{G} \pi_{gt}^2$, where π_{gt} denotes the share of residents in tract t who belong to ethnic group g, and G is the total number of ethnic groups.

[6]We therefore treat damage as missing for tracts with at least half of their area outside Houston that report zero affected units. Of the 2,002 Wave 1 respondents, 837 (41.8 percent) lack a damage value: 141 in Brazoria and 696 in tracts outside Houston's city limits. These respondents enter only specifications that use self-reported damage. OA Section A5.4 re-estimates Tables 2 and 3 using a damage measure from FEMA registrations that covers all four counties.

[7]OA Section A2 provides a detailed analysis of missingness in the variables used below.

potential confounders, such as socioeconomic conditions, rather than ethnic heterogeneity per se. Importantly, diversity is uncorrelated with most of these characteristics: correlations with income, density, and the foreign-born share are all below 0.10, and more diverse tracts are, on average, less poor ($r = -0.26$) (OA Table A5).

Figure 1: Study area: census tracts, ethnic diversity, and Harvey damage.



Notes: Left panel: ethnic diversity (fractionalization index from ACS 2015) at the census-tract level across Harris, Fort Bend, Brazoria, and Montgomery counties. Right panel: share of Wave 1 and Wave 2 respondents reporting Harvey damage, by ZIP code; grey ZIP codes have fewer than five respondents. Darker shading indicates higher values.

Results

Baseline patterns: diversity and pre-disaster coordination

Table 1 reports OLS estimates of the relationship between ethnic diversity and reported cooperation prior to Harvey with different groups in the community. The outcome is whether respondents

coordinated preparedness actions with immediate family, extended family, neighbors, or friends before the hurricane made landfall in the region. The exact survey question is: "Before Harvey reached Houston, did you coordinate plans for the approaching hurricane with others? With whom did you coordinate?" Although the wording emphasizes coordination, we interpret this behavior as a form of cooperation. These results show whether the diversity penalty found in prior work (Algan et al., 2016; Alesina and La Ferrara, 2000; Costa and Kahn, 2003; Luttmer, 2001; Miguel and Gugerty, 2005) appears in our data before the storm caused any damage. We estimate the following equation:

$$\text{Coordination}_{ij}^{k} \;=\; \alpha \;+\; \beta\,\text{Diversity}_{j} \;+\; \mathbf{X}_{ij}\boldsymbol{\gamma} \;+\; \varepsilon_{ij}^{k}, \tag{1}$$

where i denotes a respondent in census tract j, $\text{Coordination}_{ij}^{k}$ is a pre–Harvey binary measure of coordination, and k indexes social tie categories (immediate family, extended family, neighbors, or friends). All outcome variables are coded 1 when the response is affirmative and 0 otherwise. The key explanatory variable, Diversity_{j}, is the tract-level index of ethnic fractionalization. The coefficient of interest, β, captures the association between local ethnic heterogeneity and each form of coordination. $\mathbf{X}_{ij}$ is a vector of controls. Controls include indicators for partisanship, race/ethnicity, age group, educational attainment, household-income bracket, home ownership, insurance coverage, ZIP-code fixed effects, and a categorical years-in-town indicator. All models are estimated by weighted least squares using survey weights, with standard errors clustered at the census-tract level. The baseline model includes ZIP-code fixed effects and individual-level controls (party ID, age, education, race, income, years in town, home ownership, and insurance). The extended models include six tract-level controls drawn from the ACS 2015 five-year estimates: median household income, percent of renter-occupied units, percent of foreign-born residents,

population density, percent of residents below the poverty line, and percent of residents who moved in the past year.

Ethnic diversity is not correlated with coordination with family or friends, but respondents in more diverse tracts were less likely to coordinate with neighbors before Harvey (Table 1). A one-unit increase in ethnic diversity is associated with a 20.9 percentage-point (pp) decrease in the likelihood of coordinating with neighbors conditional on individual controls ($p < 0.05$); the estimate shrinks slightly to 18.7 pp once tract-level characteristics are added ($p < 0.10$). This is a large association relative to the baseline probability of coordinating with neighbors (13.3%; OA Table A2). The coefficient is positive for immediate family ($+0.08$) and most negative for neighbors (-0.19), a monotone gradient across tie strength, consistent with closer ties attenuating the diversity penalty; the sharpest contrast, immediate family versus neighbors, is significant at $p = 0.07$ (OA Table A7).

Table 1: Likelihood of coordinating prior to Hurricane Harvey.

	Immediate Family		Extended Family		Neighbors		Friends	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Ind. Controls	+ Tract Controls	Ind. Controls	+ Tract Controls	Ind. Controls	+ Tract Controls	Ind. Controls	+ Tract Controls
Diversity	-0.031	0.084	-0.163	-0.079	-0.209**	-0.187*	-0.051	-0.162
	(0.104)	(0.130)	(0.124)	(0.154)	(0.089)	(0.113)	(0.130)	(0.159)
Observations	1904	1904	1904	1904	1904	1904	1904	1904
Adj. R-squared	0.078	0.081	0.084	0.084	0.089	0.087	0.074	0.076

Notes: Results from OLS models. Each outcome is estimated twice: with individual controls only (Ind. Controls) and with the same individual controls plus six tract-level controls (+ Tract Controls): median household income, % renter-occupied, % foreign-born, population density, % below poverty, and % moved in the past year. All models include ZIP-code fixed effects, individual controls (party ID, age, education, race, income, years in town, home ownership, and insurance), and survey weights. Standard errors clustered at the census-tract level. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

No diversity penalty in interpersonal cooperation

Here, we examine whether ethnic diversity shaped patterns of interpersonal cooperation around Hurricane Harvey. We consider four measures of interpersonal cooperation: (1) whether the re-

spondent received help in the days immediately before or after Harvey;[8] (2) whether the respondent provided help to others over the same period; (3) the (log) number of volunteers known to the respondent; and (4) a cooperation index aggregating the three preceding items.[9] We estimate the following equation:

$$\text{Cooperation}_{ij}^{k} = \alpha + \beta_1 \,\text{Diversity}_j + \beta_2 \,\text{Damage}_{ij} + \beta_3 \left(\text{Diversity}_j \times \text{Damage}_{ij}\right) + \mathbf{X}_{ij}\boldsymbol{\gamma} + \varepsilon_{ij}^{k}, \tag{2}$$

where i denotes a respondent in census tract j, and $\text{Cooperation}_{ij}^{k}$ is one of the four cooperation measures listed above, indexed by k—that is, received help, provided help, the log of one plus volunteers known, or the cooperation index. Diversity_j denotes the ethnic fractionalization index, and Damage_{ij} denotes disaster exposure; we report results under three operationalizations—a self-reported binary indicator, the continuous share of housing units affected, and an indicator for damage at or above the median among respondents in damaged tracts (13.1 percent of housing units; see Table 2). The interaction term $\text{Diversity}_j \times \text{Damage}_{ij}$ allows the coefficient on diversity to differ across levels of damage. $\mathbf{X}_{ij}$ is a vector of controls as in Equation 1, with the addition of an indicator for whether the respondent evacuated before landfall. All models are estimated by weighted least squares using survey weights, with standard errors clustered at the census tract level.

[8] The received-help outcome is observed in Waves 1 and 2. Neither tract damage share nor diversity predicts missingness from this sample (OA Table A4).

[9] The exact question wording from the 2017 Wave 1 instrument is as follows. Received help: "In the days immediately before or after Hurricane Harvey, did you receive any help from family members, friends, or other people who you might know through clubs, churches or organizations you belong to?" Provided help: "In the days immediately before or after Hurricane Harvey, did you provide any help to family members, friends, or other people who you might know through clubs, churches or organizations you belong to?" Volunteers known: "How many volunteers do you know who provided help in the days immediately before or after Hurricane Harvey?" Cooperation index: rowwise mean of the standardized (z-scored) values of received help, provided help, and $\log(1 + \text{volunteers known})$, computed for respondents with non-missing values on at least two of the three items. The Wave 1 question therefore combines preparedness assistance with post-storm recovery. Table 1 shows no diversity penalty in pre-storm coordination with family or friends.

As shown in Table 2, damage is not consistently associated with cooperation. The interaction between diversity and damage is statistically indistinguishable from zero across all outcomes and the three damage specifications.[10] Overall, we find no detectable diversity penalty in interpersonal cooperation.

Table 2: Interpersonal cooperation following Hurricane Harvey.

	Received help			Provided help			log(1+Vol. known)			Coop. index		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Self-rep	Continuous	Above med	Self-rep	Continuous	Above med	Self-rep	Continuous	Above med	Self-rep	Continuous	Above med
Diversity	-0.379*	-0.009	-0.220	0.103	-0.222	-0.097	0.675	0.155	-0.105	-0.168	-0.155	-0.289
	(0.214)	(0.254)	(0.241)	(0.258)	(0.284)	(0.274)	(0.655)	(0.752)	(0.670)	(0.377)	(0.364)	(0.358)
Damage	0.116	0.695	0.094	0.152	-0.829*	-0.172	0.608*	-0.502	-0.459	0.286	-0.137	-0.126
	(0.124)	(0.433)	(0.146)	(0.143)	(0.464)	(0.200)	(0.314)	(1.026)	(0.340)	(0.224)	(0.538)	(0.216)
Diversity × Damage	0.104	-0.955	-0.012	-0.081	0.541	0.113	-0.751	-1.230	0.213	-0.061	-0.684	0.076
	(0.222)	(0.855)	(0.281)	(0.259)	(0.955)	(0.373)	(0.589)	(2.146)	(0.674)	(0.398)	(1.098)	(0.409)
Observations	2099	1194	1194	1906	1108	1108	1470	857	857	1905	1108	1108
Adj. R-squared	0.196	0.215	0.213	0.163	0.208	0.192	0.224	0.290	0.286	0.188	0.245	0.239

Notes: Results from OLS models. Each outcome is estimated under three damage measures: self-reported damage (Self-rep), administrative tract-level damage share (Continuous), and above-median administrative damage (Above med; median among damaged tracts). *Received help* (cols 1–3) is observed in both the 2017 (Wave 1) and 2018 (Wave 2) survey waves and pooled across them; the remaining three outcomes—*Provided help* (cols 4–6), *log(1+Vol. known)* (cols 7–9), and the cooperation index (cols 10–12)—are 2017 (Wave 1) items only. All specifications include individual controls (party ID, age, education, race, income, evacuation, years in town, home ownership, insurance), ZIP-code fixed effects, and six tract-level controls (median household income, % renter, % foreign-born, population density, % below poverty, % moved in the past year); the received-help specifications (cols 1–3) also include a survey-wave indicator. Survey weights included. Standard errors clustered at the census-tract level. Samples are smaller in the continuous and above-median columns because the City of Houston damage records cover only areas within city limits. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

A diversity penalty in costly public goods provision

Here, we examine a different set of measures of costly cooperation: support for recovery and adaptation policies. We construct an index of policy support equal to the average of ten binary survey questions on support for specific policy proposals to protect the Houston area from severe weather events:[11]

[10] OA Table A8 reports the confidence intervals for these interactions and their magnitudes in percentage points.

[11] The items are: (i) a program to buy homes in areas that have repeatedly flooded with local, state, and federal moneys; (ii) construction of a new reservoir to protect the western portion of the Houston area; (iii) greater restrictions on construction in flood plains; (iv) establishment of a regional flood agency with taxing authority to plan for the prevention of regional flooding; (v) denying federally financed flood insurance to homeowners whose homes have flooded three or more times since 2001; (vi) not allowing homes that have flooded three or more times since 2001 to

$$\text{Policy support}_{ij} = \frac{1}{10}\left(\text{policy}_{ij}^{(\text{i})} + \text{policy}_{ij}^{(\text{ii})} + \cdots + \text{policy}_{ij}^{(\text{x})}\right), \quad \text{with policy}_{ij}^{(\cdot)} \in \{0, 1\}.$$

We estimate the following equation:

$$\begin{aligned} \text{Policy support}_{ij} \;=\; & \alpha \;+\; \beta_1 \,\text{Diversity}_j \;+\; \beta_2 \,\text{Damage}_{ij} \;+\; \beta_3 \left(\text{Diversity}_j \times \text{Damage}_{ij}\right) \\ & +\; \mathbf{X}_{ij}\boldsymbol{\gamma} \;+\; \varepsilon_{ij}, \end{aligned} \tag{3}$$

where i indexes respondents in census tract j, the dependent variable Policy support$_{ij}$ is the index described above, Diversity$_j$ is the ethnic fractionalization index, and Damage$_{ij}$ denotes disaster exposure, which we operationalize using the same three measures as above: self-reported damage, the continuous tract-level share of housing units affected, and the above-median indicator. The interaction term Diversity$_j \times$Damage$_{ij}$ allows the association between diversity and policy support to vary with damage exposure. $\mathbf{X}_{ij}$ is a vector of controls as in the earlier models, including an indicator for evacuation before landfall. All models are estimated by weighted least squares using survey weights, with standard errors clustered at the census-tract level.

Greater damage is associated with higher policy support across all three specifications, with the association reaching statistical significance for self-reported damage (Table 3). This result is consistent with work on how disaster exposure affects support for climate policies (Arias and Blair, 2024; Bechtel and Mannino, 2023; Bergquist and Warshaw, 2019). The interaction between damage and diversity is negative in all three columns, significant at $p < 0.05$ under self-reported

be rebuilt by buying out these homeowners with local and federal moneys; (vii) requiring sellers of homes to fully disclose prior flood damage to their homes and prior flooding in the surrounding neighborhood; (viii) preventing development/construction on native prairies and wetlands in western and northwestern portions of Harris County; (ix) requiring government compensation for homes that are flooded due to the release of water from local reservoirs; and (x) new building codes that require homes built in flood-prone areas to be elevated/raised to avoid flooding.

damage and at $p < 0.10$ under the two administrative measures. Under the self-reported specification, the marginal effect of damage on policy support is approximately $+7$ pp in lower-diversity tracts ($Diversity_j = 0.25$) and -3 pp in higher-diversity tracts ($Diversity_j = 0.75$)—a difference of roughly 10 pp, about 13% of the baseline mean of 78.4%. The continuous damage specification yields the same pattern: each additional percentage point of tract-level damage is associated with a change in policy support about 0.35 pp smaller at $Diversity_j = 0.75$ than at $Diversity_j = 0.25$ ($p < 0.10$).

Across the ten items, the interaction is negative for most, and the items that reach statistical significance vary with the damage measure: under the continuous measure they are restrictions on flood-plain construction and the denial of federal flood insurance to repeat claimants (both $p < 0.05$), whereas under self-reported damage it is the buyout program (OA Table A10). No consistent split between spending and regulatory items emerges; the "Alternative explanations" subsection below tests this and the time-horizon account directly.[12]

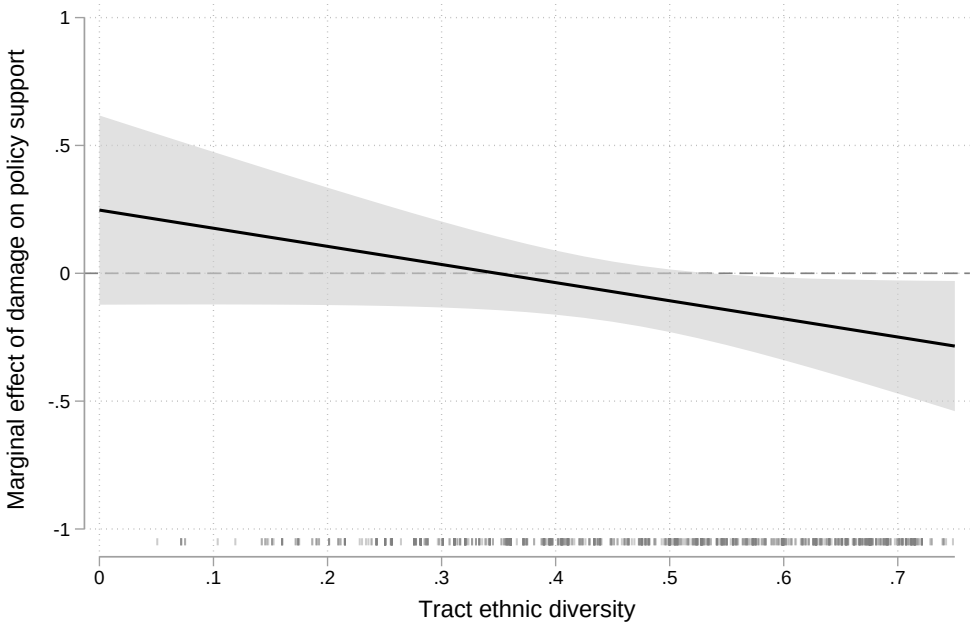
[12] A potential alternative explanation is that damaged residents prioritize private goods over costly public goods due to shortsightedness (Visconti, 2022b). However, note that this would imply a diversity penalty that is constant in damage. Instead, the monitoring argument predicts that the penalty increases with damage—the pattern we observe.

Table 3: Support for disaster recovery and adaptation policies following Hurricane Harvey.

	Support for Policies (Index)		
	(1)	(2)	(3)
	Self-reported	Continuous	Above median
Diversity	0.083	0.018	-0.025
	(0.084)	(0.113)	(0.096)
Damage	0.118***	0.247	0.074
	(0.044)	(0.188)	(0.065)
Diversity × Damage	-0.200**	-0.709*	-0.213*
	(0.080)	(0.391)	(0.122)
Observations	1905	1107	1107
Adj. R-squared	0.104	0.174	0.173

Notes: Results from OLS models. Three damage specifications: self-reported (Self-reported), administrative tract damage share (Continuous), and above-median administrative damage (Above median; median among damaged tracts). All specifications include individual controls (party ID, age, education, race, income, evacuation, years in town, home ownership, and insurance), ZIP-code fixed effects, and six tract-level controls (median household income, % renter, % foreign-born, population density, % below poverty, % moved in the past year). Survey weights included. Standard errors clustered at the census-tract level. Samples are smaller in the continuous and above-median columns because the City of Houston damage records cover only areas within city limits. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Figure 2: Marginal effect of damage on policy support across tract diversity.



Notes: Marginal effect of tract damage share on the policy-support index as a function of tract ethnic diversity, estimated from the continuous-damage specification in Table 3. The black line plots the marginal effect of damage at each level of diversity; the shaded area denotes the 95% confidence interval (clustered at the tract). Tick marks at the bottom show the distribution of tract diversity in the estimation sample. The marginal effect of damage is positive but not statistically significant in homogeneous tracts ($+0.25$ at Diversity $= 0$, $p = 0.19$), declines with diversity, crosses zero near Diversity ≈ 0.35, and becomes significantly negative past Diversity ≈ 0.55, reaching about -0.29 at Diversity $= 0.75$. $N = 1{,}107$.

Figure 2 plots how the association between damage and policy support varies with tract diversity. In the most homogeneous tracts, an additional unit of tract damage share is associated with a 0.25 increase in the policy-support index (not significant, $p = 0.19$). The marginal effect declines monotonically as diversity increases, crosses zero around $Diversity_j \approx 0.35$, and becomes significantly negative beyond $Diversity_j \approx 0.55$, reaching -0.29 at the upper end of the observed diversity range.[13]

[13]The pattern does not depend on the linear form of the interaction: estimating the damage slope separately within each quartile of diversity yields a marginal effect indistinguishable from zero in the bottom three quartiles and -0.53 in the top quartile (OA Table A9).

To test whether the diversity penalty grows with the size of the required contribution, we use a conjoint experiment from the third survey wave that randomizes the tax cost of recovery proposals. Respondents were presented with pairs of policy proposals that varied randomly along three dimensions: (1) project type (road infrastructure, loans to rebuild housing, or infrastructure to prevent new flooding), (2) the geographic area covered by the policy (poor residential areas, industrial areas, commercial areas, or Houston-wide), and (3) the level of tax increase. The three project types reflect policy debates that followed Harvey: Harris County voters approved a $2.5 billion flood-control bond in August 2018 that funds a home buyout program, the Houston City Council tightened flood-plain elevation rules that April, and Texas expanded flood-disclosure requirements for home sellers in 2019.[14] Choosing the higher-tax proposal reflects willingness to make a costly contribution to a public good whose benefits also accrue to others, making the choice a measure of costly cooperation rather than simple policy preference. The original attribute randomized sales and property tax schedules, each displayed with its annual dollar cost: $200, $400, or $600 per year for a $225,000 house or a household income of $60,000 (Figure 3).[15]

[14] *Texas Tribune*, August 25, 2018; April 4, 2018; August 22, 2019.

[15] Throughout, the policy conjoint analyses are reported on the Houston-only Wave 3 sample. The full Wave 3 sample, which includes respondents from the four-county study area, is reported in OA Table A13 as a robustness check.

Figure 3: Policy conjoint experiment: sample scenario.

In each of the following pages, you will be shown profiles from **two different** proposals associated with generating funds for recovery/prevention of natural disasters and your willingness to pay for funding of those programs. For each pair, read each profile carefully and indicate your response. Please consider each pair of profiles independently of the pairs listed on other pages.

Attribute	**Proposal A**	**Proposal B**
Tax Option	1% increase in sales tax (extra $200 tax per year for a household income of $60,000)	0.2% increase in property tax (extra $400 tax per year for a $225,000 house)
To be used for	Build infrastructure to prevent new flooding	Recover road infrastructure affected by Harvey
Project	Poor residential areas	Throughout the Houston area

Which of these two proposals do you prefer?

- ○ **Proposal A**
- ○ **Proposal B**

The conjoint presented seven tax schedules, spanning from no increase to a 3 percent sales-tax increase and a 0.3 percent property-tax increase. To test whether respondents are willing to endorse the more costly policy, the tax attribute—regardless of the tax base (sales or property)—was recoded as a binary variable that takes the value 1 when the profile shows the higher of the two tax levels in the pair and 0 when it shows the lower tax level; profiles in which both options display the same tax level are coded as missing. After this transformation, 3,700 profile choices

are classified as Higher Tax Increase and 3,700 as Lower Tax Increase, while 1,120 tied profiles are excluded from the analysis.[16] We estimate the following linear probability mixed model:

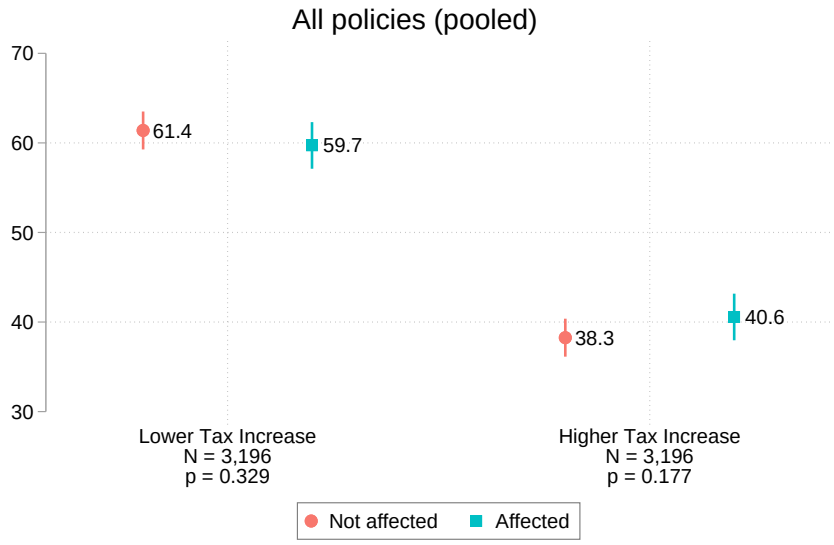
$$\begin{aligned}\text{Choice}_{itp} &= \alpha + b_1\,\text{Use}_p + b_2\,\text{Projects}_p + b_3\,\text{HighTax}_{tp} + b_4\,\text{Affected}_i + b_5\big(\text{Use}_p \times \text{Affected}_i\big) \\ &\quad + b_6\big(\text{Projects}_p \times \text{Affected}_i\big) + b_7\big(\text{HighTax}_{tp} \times \text{Affected}_i\big) + u_t + v_i + \varepsilon_{itp},\end{aligned} \tag{4}$$

where the dependent variable Choice_{itp} is a binary variable capturing whether profile p evaluated by respondent i in paired trial t was selected. The attribute Use_p denotes the policy type, and Projects_p refers to the target area of the project. The binary variable HighTax_{tp} equals 1 when the profile carries the higher of the two tax levels in the pair and 0 otherwise. We interact each attribute with Affected_i, an indicator of whether the respondent reported being personally affected by Harvey. The random intercepts u_t and v_i account for unobserved heterogeneity at the trial and respondent levels.

Figure 4 presents the results and OA Table A13 reports the full set of estimates. In general, respondents were more likely to support proposals that involved lower rather than higher tax increases. Among non-affected respondents, the high-tax penalty—the coefficient on HighTax_{tp}—is approximately -23 pp on the 0–100 choice probability scale: non-affected respondents are roughly 23 pp less likely to choose the higher-tax proposal than the lower-tax proposal. This aversion is weaker among affected respondents: the coefficient on $\text{Affected}_i \times \text{HighTax}_{tp}$ is $+4.2$ pp ($p < 0.10$) in the Houston-only sample and $+8.0$ pp ($p < 0.01$) in the full sample as a robustness check (OA Table A13). This coefficient corresponds to the change in the vertical gap between affected and non-affected respondents across the two tax conditions in Figure 4.

[16] 304 observations were excluded due to missingness in the damage survey question. Tied pairs contain no within-pair variation in the tax attribute and therefore do not identify the tax effect. In the Houston-only sample of affected respondents used in Equation 5 below, 186 of 1,460 pairs are tied, leaving 2,548 profile evaluations.

Figure 4: Predicted margins of choosing high versus low tax increases following Hurricane Harvey conditional on being affected by the storm.



Notes: Points plot predicted margins of choosing the proposed policy option for respondents affected by Hurricane Harvey (teal squares) and not affected (salmon circles), conditional on the size of the required tax increase (lower vs. higher). Labels below each tax condition report the number of profile evaluations (N) and the p-value for the affected–unaffected difference at that tax level. Vertical bars denote 95% confidence intervals. Houston-only Wave 3 sample. The decomposition by policy type appears in OA Figure A3, alongside the underlying coefficients in OA Table A13.

Equation (5) re-estimates the conjoint model for the 365 respondents who reported being affected by Hurricane Harvey in the Houston-only Wave 3 sample (38.5 percent of the sample).

$$\begin{aligned} \text{Choice}_{itp} &= \alpha + \beta_1 \,\text{Use}_p + \beta_2 \,\text{Projects}_p + \beta_3 \,\text{HighTax}_{tp} + \beta_4 \,\text{HighDiversity}_i \\ &\quad + \beta_5 \,\text{Use}_p \times \text{HighDiversity}_i + \beta_6 \,\text{Projects}_p \times \text{HighDiversity}_i \\ &\quad + \beta_7 \,\text{HighTax}_{tp} \times \text{HighDiversity}_i + u_t + v_i + \varepsilon_{itp}, \end{aligned} \tag{5}$$

where Choice_{itp} equals 1 when profile p is selected in paired trial t by respondent i. The covariates Use_p, Projects_p, and HighTax_{tp} retain their earlier definitions. The binary variable HighDiversity_i

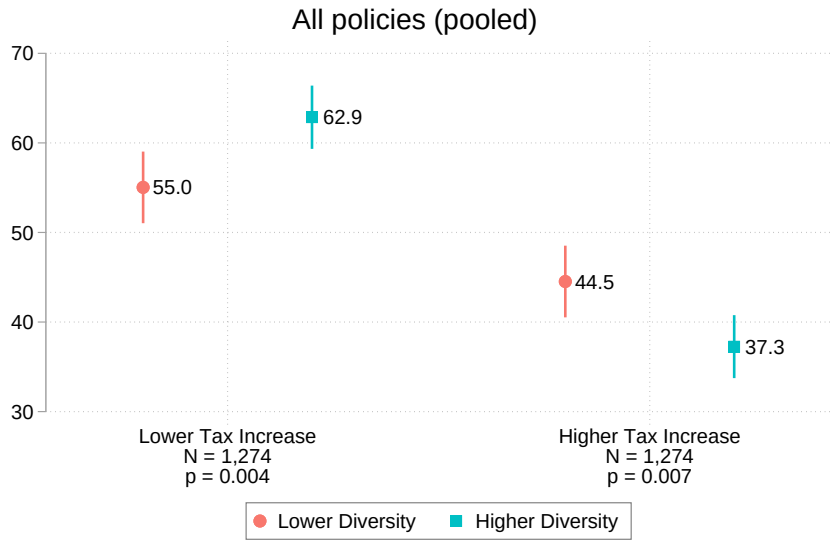
takes the value 1 when the ZIP-code ethnic fractionalization index for respondent i's ZIP code exceeds the sample mean and 0 otherwise.[17]

Across policies, affected respondents living in higher-diversity areas are less likely to choose the high-tax proposal and more likely to choose the low-tax proposal (Figure 5). OA Table A13 reports the full set of estimates. Among lower-diversity affected respondents, the high-tax penalty—the coefficient on HighTax_{tp}—is approximately -10 to -11 pp on the 0–100 choice probability scale: an affected respondent in a lower-diversity area is 10–11 pp less likely to choose the higher-tax proposal than the lower-tax proposal. The diversity–tax penalty—the coefficient on $\text{HighTax}_{tp} \times \text{HighDiversity}_i$—is an additional -15.1 pp ($p < 0.01$) in the Houston-only sample and -9.4 pp ($p < 0.01$) in the full sample as a robustness check: respondents in higher-diversity areas are much more averse to higher-tax proposals. This coefficient corresponds to the change in the vertical gap between higher- and lower-diversity respondents across the two tax conditions in Figure 5.[18]

[17] Because Wave 3 collected respondents' ZIP codes rather than street addresses, the diversity index and the additional controls in the conjoint analyses are constructed at the ZIP-code level, rather than at the census-tract level used in the observational analyses (Tables 1–3). See OA Section A6.

[18] The tax attribute is randomized within respondent, so the interaction cannot reflect respondent- or area-level differences in tax aversion; it is unchanged when respondent partisanship is added as a control or interacted with the tax attribute (OA Table A14).

Figure 5: Predicted margins of choosing high versus low tax increases following Hurricane Harvey for affected respondents conditional on ethnic diversity.



Notes: Points plot predicted margins of choosing the proposed policy option for respondents living in lower-diversity areas (below-mean ethnic diversity, salmon circles) and higher-diversity areas (above-mean, teal squares), conditional on the size of the required tax increase (lower vs. higher). Labels beneath each tax condition report the number of profile evaluations (N) and the p-value for the higher–lower diversity difference at that tax level. Vertical bars denote 95% confidence intervals. Houston-only Wave 3 sample, affected respondents only. The decomposition by policy type appears in OA Figure A4, alongside the underlying coefficients in OA Table A13.

A second conjoint experiment in Wave 3 asked respondents to choose between two hypothetical neighbors as partners in providing information and help during future flooding (OA Section A7). The profiles randomized gender, race, partisanship, religion, and civic-association membership. Shared partisanship and shared religion increase the probability of choosing a profile by about 25 and 17 percentage points, respectively, with similar effects across diversity levels. The effect of shared civic-association membership is larger in higher-diversity ZIP codes than in lower-diversity ZIP codes—9.2 versus 4.5 percentage points—a pattern consistent with associations serving as observable signals where ascriptive cues are less informative. The cross-diversity difference, however, is not statistically significant ($p = 0.31$), so we treat this result as suggestive evidence consistent with our monitoring argument.

Alternative explanations

Support for recovery policies differs from interpersonal help in ways other than the feasibility of monitoring. Two differences could in principle produce the pattern above: fiscal cost and time horizon. Tables 4 and 5 re-estimate Equation 3 on subsets of the policy index chosen to separate each alternative from the monitoring account.

First, recovery policies cost money and interpersonal help usually does not. If diversity depressed support because policies require visible fiscal costs, the penalty should concentrate on the five items that require public spending or taxing authority—buyouts, a new reservoir, a regional flood agency with taxing authority, buyouts in place of rebuilding, and compensation for reservoir releases—and disappear on the five items that impose regulatory or private costs. It does not. The interaction is -0.71 ($p = 0.09$) on the regulatory items and -0.62 ($p = 0.19$) on the spending items, and the two estimates do not differ ($p = 0.82$; Table 4).

Table 4: Support for recovery policies by fiscal visibility of the policy items.

	Full index (10 items) (1)	Spending items (5) (2)	Regulatory items (5) (3)
Self-reported damage			
Diversity × Damage	-0.200**	-0.232**	-0.145*
	(0.080)	(0.114)	(0.084)
Observations	1,905	1,899	1,903
Wald test H_0: (2) = (3)	$\chi^2(1) = 0.67$, $p = 0.41$		
Continuous damage share			
Diversity × Damage	-0.709*	-0.615	-0.711*
	(0.391)	(0.470)	(0.423)
Observations	1,107	1,102	1,105
Wald test H_0: (2) = (3)	$\chi^2(1) = 0.05$, $p = 0.82$		
Above-median damage			
Diversity × Damage	-0.213*	-0.229	-0.161
	(0.122)	(0.163)	(0.121)
Observations	1,107	1,102	1,105
Wald test H_0: (2) = (3)	$\chi^2(1) = 0.23$, $p = 0.63$		
Individual controls	Yes	Yes	Yes
Tract controls	Yes	Yes	Yes
ZIP-code fixed effects	Yes	Yes	Yes

Notes: OLS estimates. The dependent variable is the share of policy items the respondent supports, scaled 0–1. Column 1 uses all ten items; column 2 the five items that require government outlays or taxing authority (buyout program; new reservoir; regional flood agency with taxing authority; buyouts of repeatedly flooded homes in lieu of rebuilding; compensation for reservoir-release flooding); column 3 the five items that impose private or regulatory costs with no public spending (flood-plain construction restrictions; denial of federal flood insurance to repeat claimants; seller disclosure; prohibition of development on prairies and wetlands; elevation building codes). Each panel uses one of the three damage measures of Table 3: self-reported damage (binary), the share of housing units affected in the respondent's census tract (continuous), and an indicator for damage at or above the median among damaged tracts. All specifications include individual controls (party identification, race, age, education, income, evacuation, years in town, home ownership, insurance), six ACS 2015 tract controls, and ZIP-code fixed effects, and are weighted by survey weights; standard errors, in parentheses, are clustered at the census-tract level. The Wald test stacks columns 2 and 3 by seemingly unrelated estimation and tests equality of the Diversity × Damage coefficients. Samples are smaller in the continuous and above-median panels because the City of Houston damage records cover only areas within city limits. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Second, many recovery policies are forward-looking and help is immediate. If the diversity penalty reflected reluctance to sacrifice for distant benefits across group lines, it should concentrate on the eight mitigation items and be weaker on the two items that address damage already incurred.

The estimate is at least as large on the response items (-0.89, $p = 0.18$) as on the mitigation items (-0.67, $p = 0.09$), and the two estimates do not differ ($p = 0.68$; Table 5). The policy conjoint shows the same ordering: the diversity–tax interaction is -16.8 percentage points for road recovery and housing loans and -11.1 for flood-prevention infrastructure (OA Table A16). In neither case does the penalty vanish where the alternative predicts it should; neither fiscal cost nor time horizon accounts for the pattern we document.

Table 5: Support for recovery policies by time horizon.

	Full index (10 items) (1)	Response items (2) (2)	Mitigation items (8) (3)
Self-reported damage			
Diversity × Damage	-0.200**	-0.350**	-0.168**
	(0.080)	(0.143)	(0.082)
Observations	1,905	1,866	1,904
Wald test H_0: (2) = (3)	$\chi^2(1)$ = 2.04, p = 0.15		
Continuous damage share			
Diversity × Damage	-0.709*	-0.893	-0.670*
	(0.391)	(0.660)	(0.392)
Observations	1,107	1,091	1,106
Wald test H_0: (2) = (3)	$\chi^2(1)$ = 0.17, p = 0.68		
Above-median damage			
Diversity × Damage	-0.213*	-0.218	-0.213*
	(0.122)	(0.227)	(0.118)
Observations	1,107	1,091	1,106
Wald test H_0: (2) = (3)	$\chi^2(1)$ = 0.00, p = 0.98		
Individual controls	Yes	Yes	Yes
Tract controls	Yes	Yes	Yes
ZIP-code fixed effects	Yes	Yes	Yes

Notes: OLS estimates. The dependent variable is the share of policy items the respondent supports, scaled 0–1. Column 1 uses all ten items; column 2 the two backward-looking response items (the buyout program for repeatedly flooded homes; compensation for homes flooded by reservoir releases); column 3 the eight forward-looking mitigation items. Each panel uses one of the three damage measures of Table 3: self-reported damage (binary), the share of housing units affected in the respondent's census tract (continuous), and an indicator for damage at or above the median among damaged tracts. All specifications include individual controls (party identification, race, age, education, income, evacuation, years in town, home ownership, insurance), six ACS 2015 tract controls, and ZIP-code fixed effects, and are weighted by survey weights; standard errors, in parentheses, are clustered at the census-tract level. The Wald test stacks columns 2 and 3 by seemingly unrelated estimation and tests equality of the Diversity × Damage coefficients. Samples are smaller in the continuous and above-median panels because the City of Houston damage records cover only areas within city limits. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$

Conclusion

We examined how disaster exposure and ethnic diversity relate to different forms of cooperation. Our main finding is that cooperation does not respond uniformly to shocks: the relationship be-

tween disaster exposure and cooperation varies across domains, depending on the feasibility of monitoring.

Using observational and experimental data collected around Hurricane Harvey, we uncovered two patterns. First, we found no diversity penalty in interpersonal, face-to-face cooperation following the disaster, where actions are directly observable. Second, for costly forms of cooperation in the provision of public goods, diversity moderates responses to shocks: while exposure to damage is associated with greater support for disaster recovery and adaptation policies, this relationship weakens and reverses in more diverse areas.[19]

Together, our findings suggest that diversity is not uniformly associated with lower cooperation; rather, it shapes different forms of cooperation in distinct ways. This distinction matters for interpreting the role of race and diversity in disaster politics. While recent work shows that individual racial attitudes shape disaster-spending preferences (Gilens et al., 2025), our results point to a distinct contextual logic: local ethnic diversity reduces support for disaster recovery and adaptation policies when those policies require contributions that are difficult to observe and sanction. Where contributions are observable, we do not detect a consistent diversity penalty; where contributions are difficult to monitor and vulnerable to free-riding, cooperation declines with diversity.

More broadly, the results suggest that institutional features that make contributions more observable may shape how diversity affects cooperation. In the context of disaster recovery and climate adaptation, policies that rely on contributions that are difficult to monitor are likely to be more sensitive to social characteristics that shape the scope for free-riding. Civic-association membership may serve as one such observable signal in more diverse settings. Future work can assess whether these patterns extend across contexts and types of shocks.

[19] An alternative interpretation is that diversity penalizes weak-tie cooperation but not strong-tie cooperation. However, this account is difficult to reconcile with the finding that the diversity penalty increases with the stakes of cooperation. Under the monitoring argument, larger contributions increase incentives for free-riding, so the diversity penalty should increase with the stakes, as in the high-tax options in the conjoint analysis (OA Table A13). A tie-strength account, by contrast, predicts a uniform penalty regardless of the size of the contribution.

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Conflict of interest. The authors of this manuscript have no competing interests or conflicts of interest to declare.

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